

Lesson 7: Infinite Limits and Limits at Infinity

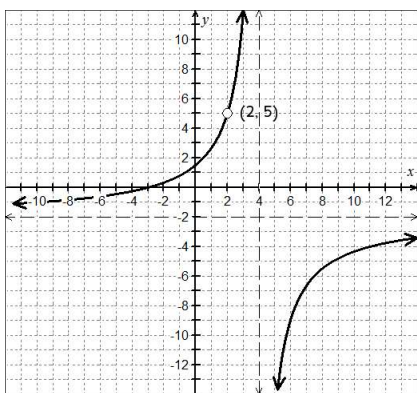
In our first lesson on Understanding Limits you were confronted with these two situations. Here we begin to **compare and contrast** the behavior of functions as they **approach infinity**, as well as, functions that **tend toward infinity** in certain circumstances. Let's go...

Topic 1.14: Connecting Infinite Limits and Vertical Asymptotes

Topic 1.15: Connecting Limits at Infinity and Horizontal Asymptotes

EX #1: Use the graphs of $f(x)$ and $g(x)$ shown below to compare and contrast behaviors involving infinity. Then, write your discovery and definition.

$$f(x) = \frac{-2x^2 - 2x + 12}{x^2 - 6x + 8}$$



$$\lim_{x \rightarrow 4^-} f(x) =$$

$$\lim_{x \rightarrow 1^-} g(x) =$$

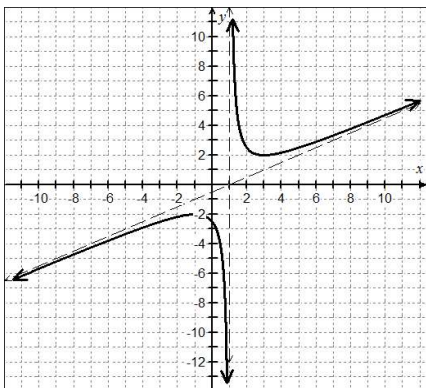
$$\lim_{x \rightarrow 4^+} f(x) =$$

$$\lim_{x \rightarrow 1^+} g(x) =$$

Your Discovery:

An Infinite Limit is:

$$g(x) = \frac{x^2 - 2x + 5}{2x - 2}$$



$$\lim_{x \rightarrow -\infty} f(x) =$$

$$\lim_{x \rightarrow -\infty} g(x) =$$

$$\lim_{x \rightarrow \infty} f(x) =$$

$$\lim_{x \rightarrow \infty} g(x) =$$

Your Discovery:

A Limit at Infinity is:

In Pre-Calculus you learned some basic truths about rational functions.

1. When a factor cancelled from the denominator a _____ occurred.
2. When a factor would not cancel from the denominator a _____ occurred.

EX #2: Use the previous equations to find the one-sided limits analytically.

A. $\lim_{x \rightarrow 4^-} \frac{-2x^2 - 2x + 12}{x^2 - 6x + 8}$

B. $\lim_{x \rightarrow 4^+} \frac{-2x^2 - 2x + 12}{x^2 - 6x + 8}$

C. $\lim_{x \rightarrow 1^-} \frac{x^2 - 2x + 5}{2x - 2}$

D. $\lim_{x \rightarrow 1^+} \frac{x^2 - 2x + 5}{2x - 2}$

Definition and Justification of Vertical Asymptotes

Case #1:

$$h(c) = \frac{\text{non-zero}}{\text{zero}}$$

$x = c$ is:

Case #2:

$$h(c) = \frac{\text{zero}}{\text{zero}}$$

$x = c$ is:

IN CALCULUS, YOU MUST USE NEW LANGUAGE IN ORDER TO JUSTIFY!

LIMIT DEFINITION (JUSTIFICATION) OF A VERTICAL ASYMPTOTE

Topic 1.13: Removing Discontinuities

EX #3: Use the function below to find any vertical asymptote(s) that exist. Justify your answer using limits.

$$h(x) = \frac{2x^2 + 9x - 5}{x^2 + 3x - 10}$$

Limits at Infinity

Next, we will explore **limits at infinity** in order to differentiate between the two conditions. Recall the lessons from Pre-Calculus related to analyzing the **end behavior of functions**. In the exercise below, use this prior knowledge to find each limit at infinity.

EX #4: Find each limit at infinity, explain your thinking.

A. $\lim_{x \rightarrow \infty} (x^2 - 4)(x^2 + 3)$

B. $\lim_{x \rightarrow -\infty} (5x^3 - 2x + 4)$

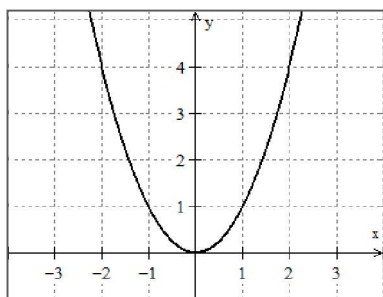
C. $\lim_{x \rightarrow \infty} \frac{3x^2 - 4}{x^2 + 1}$

D. $\lim_{x \rightarrow -\infty} \frac{5x - 2}{x^2 + 1}$

EX #5: There are only four possible outcomes when you explore behavior to the extreme right or left.

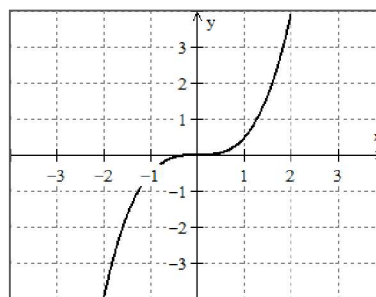
A. The curve can increase without bound.

$$\lim_{x \rightarrow \infty} f(x) =$$



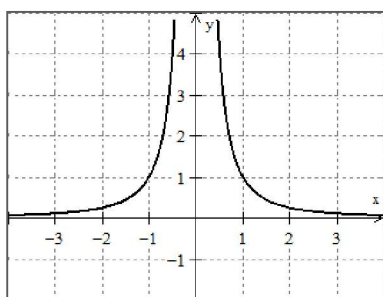
B. The curve can decrease without bound.

$$\lim_{x \rightarrow -\infty} f(x) =$$



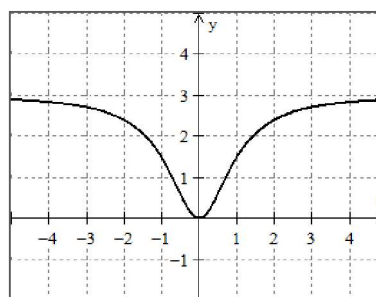
C. The curve can become asymptotic to the x-axis.

$$\lim_{x \rightarrow \infty} f(x) =$$



D. The curve can become asymptotic to a specific y-value.

$$\lim_{x \rightarrow -\infty} f(x) =$$



Calculus Knowledge for Asymptotes

Revisiting the rules for finding potential horizontal asymptotes for rational functions from PreCalculus, you can use the idea of a limit and calculus to see why those rules hold true.

1. If degree of numerator is less than degree of denominator (bottom heavy), then limit is zero.

$$\lim_{x \rightarrow \pm\infty} f(x) = 0$$

2. If degree of numerator equals degree of denominator (powers equal), then limit is the ratio of coefficients of the highest degree.

$$\lim_{x \rightarrow \pm\infty} f(x) = \frac{\text{coefficient of numerator's highest power}}{\text{coefficient of denominator's highest power}}$$

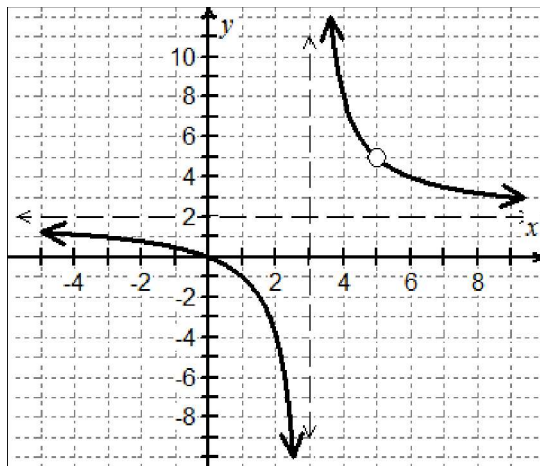
3. If degree of numerator is greater than degree of denominator (top heavy*), then limit does not exist.

$$\lim_{x \rightarrow \pm\infty} f(x) = \pm\infty$$

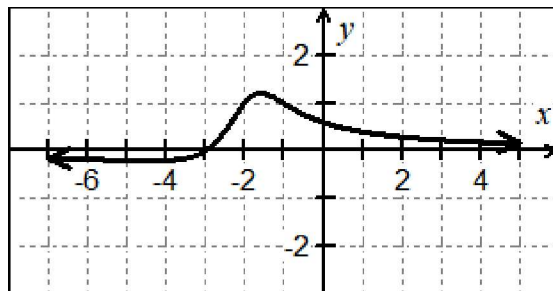
Topic 1.9: Connecting Multiple Representations of Limits

EX #6: Divide every term in the rational expression by the *highest power of x that appears in the denominator*. Then, apply the Properties of Limits to evaluate each “piece” to find the limit at infinity, end behavior.

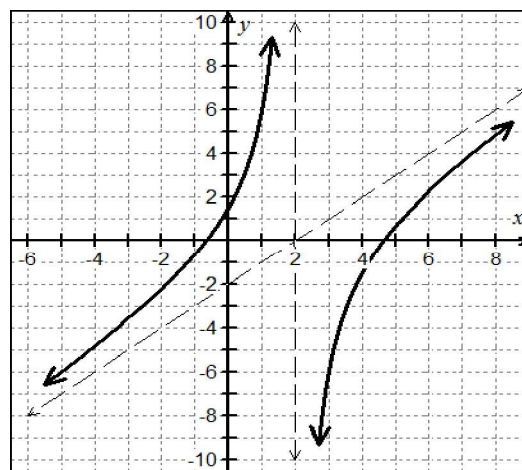
A.
$$\lim_{x \rightarrow \infty} \frac{2x^2 - 10x}{x^2 - 8x + 15}$$



B.
$$\lim_{x \rightarrow \infty} \frac{x + 3}{x^2 + 4x + 5}$$



C.
$$\lim_{x \rightarrow \infty} \frac{x^2 - 4x - 3}{x - 2}$$



EX #7: Summarize and discuss characteristics and end behavior at horizontal asymptotes and slant asymptotes based on your observations in EX #6.

Functions with Horizontal Asymptotes

$$\lim_{x \rightarrow \infty} f(x) =$$

$$\lim_{x \rightarrow -\infty} f(x) =$$

Functions with Slant Asymptotes

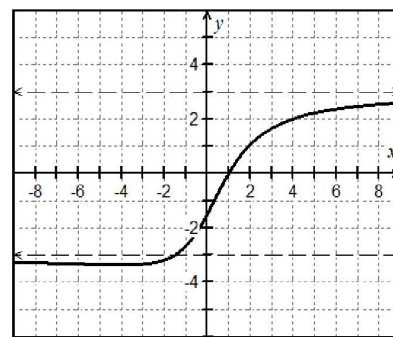
$$\lim_{x \rightarrow \infty} f(x) =$$

$$\lim_{x \rightarrow -\infty} f(x) =$$

IN CALCULUS, YOU MUST USE NEW LANGUAGE IN ORDER TO JUSTIFY!

LIMIT DEFINITION (JUSTIFICATION) OF A HORIZONTAL ASYMPTOTE

EX #8: CHALLENGE! Use algebraic techniques to find the limits for $g(x) = \frac{3x - 3}{\sqrt{x^2 + 4}}$, whose graph is shown.



$$\lim_{x \rightarrow -\infty} \frac{3x - 3}{\sqrt{x^2 + 4}} =$$

$$\lim_{x \rightarrow \infty} \frac{3x - 3}{\sqrt{x^2 + 4}} =$$